

Vector Calculus

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Green's Theorem

Tangent form or Circulation-Curl form

$$\oint_C Mdx + Ndy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA$$

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$$\oint_C F \cdot dr = \iint_R (\nabla \times F) \cdot k dA$$

- ▶ C is a simple, closed, smooth curve in counterclockwise direction
- ▶ R is the region enclosed by C
- ▶ dA is area element
- ▶ dr is tangential length

Stokes Theorem

The circulation of $F = M\mathbf{i} + N\mathbf{j} + P\mathbf{k}$ around the boundary C of an oriented surface S in the direction counterclockwise with respect to the surface's unit normal vector $\mathbf{n}$ equals the integral of $\nabla \times F \cdot \mathbf{n}$ over S

$$\oint_C F \cdot d\mathbf{r} = \iint_S \nabla \times F \cdot \mathbf{n} \, d\sigma$$

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$$\nabla \times F = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 & 2x & z^2 \end{vmatrix}$$

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Exercise

Stoke's Theorem

Use Stoke's theorem to calculate the flux of the curl of the field F across the surface S in the direction of the outward unit normal n .

1. $F = 2zi + 3xj + 5yk$

$$S : z + x^2 + y^2 = 4$$

$$12\pi$$

2. $F = 2zi + 3xj + 5yk$

$$S : \mathbf{r}(r, \theta) = (r \cos \theta)i + (r \sin \theta)j + (4 - r^2)k$$

$$0 \leq r \leq 2, 0 \leq \theta \leq 2\pi$$

3. $F = x^2yi + 2y^3zj + 3zk$

$$S : \mathbf{r}(r, \theta) = (r \cos \theta)i + (r \sin \theta)j + rk$$

$$0 \leq r \leq 1, 0 \leq \theta \leq 2\pi$$

$$\frac{-\pi}{4}$$

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(Normal form or Flux-Divergence form)

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$$\oint_C F \cdot n \, ds = \iint_S \nabla \cdot F \, dA$$

- ▶ C is a simple, closed, smooth curve
- ▶ R is the region enclosed by C
- ▶ dA is area element
- ▶ ds is length element

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$$\iint_S F \cdot n \, d\sigma = \iiint_D \nabla \cdot F \, dV$$

- ▶ S is a simple, closed, oriented surface.
- ▶ D is solid region bounded by S
- ▶ $d\sigma$ surface area element
- ▶ dV is volume element

The Divergence Theorem

The flux of a vector field $F = M\mathbf{i} + N\mathbf{j} + P\mathbf{k}$ across a closed oriented surface S in the direction of the surface's outward unit normal field n equals the integral of $\nabla \cdot F$ (divergence of F) over the region D enclosed by the surface:

$$\iint_S F \cdot n \, d\sigma = \iiint_D \nabla \cdot F \, dV.$$

$$F = yi + xyi - zk$$

D : The region inside the solid cylinder $x^2 + y^2 \leq 4$ between the plane $z = 0$ and the paraboloid $z = x^2 + y^2$

$$\nabla \cdot F = 0 + x - 1 = x - 1$$

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$$\begin{aligned} \iiint_D \nabla \cdot F \, dV &= \int_0^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_0^{x^2+y^2} (x-1) \, dz \, dy \, dx \\ &= \int_0^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} (x-1)(x^2+y^2) \, dy \, dx \\ &= \int_0^2 (x-1) \left[x^2 y + \frac{y^3}{3} \right]_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \\ &= \int_0^2 (x-1) \left(2x^2 \sqrt{4-x^2} + \frac{2}{3} (4-x)^2 \sqrt{4-x^2} \right) dx \\ &= \frac{1}{3} \int_0^2 (x-1) \sqrt{4-x^2} [6x^2 + 2(16-8x+8x^2)] dx \\ &= \frac{1}{3} \int_0^2 (x-1) \sqrt{4-x^2} [8x^2 - 8x + 16] dx \\ &= -16\pi \end{aligned}$$

Exercise

Divergence theorem

Use divergence theorem to calculate outward flux

1. $F = (y - x)\mathbf{i} + (z - y)\mathbf{j} + (y - x)\mathbf{k}$

D : The cube bounded by the planes $x \pm 1$, $y \pm 1$ and $z \pm 1$.

-16

2. $F = x^2\mathbf{i} - 2xy\mathbf{j} + 3xz\mathbf{k}$

D : The region cut from the first octant by the sphere $x^2 + y^2 + z^2 = 4$

3π

- ▶ F is conservative, F is irrotational $\implies$ Circulation = 0
- ▶ F is incompressible, $\nabla \cdot F$ is 0 $\implies$ Flux = 0

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$$= \int_{[a,b]} \nabla \cdot F dx$$

Integral of the differential operator acting on a field over a region equal the sum of (or integral of) field components appropriate to the operator on the boundary of the region